

WIND WAVE GENERATION

- Waves are generated by the tangential (shear) and the normal (pressure) stresses exerted by the wind over the sea surface

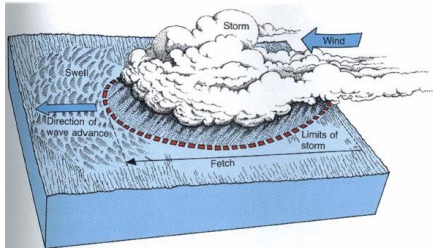


Table 1.1 A selection of information from the Beaufort Wind Scale.

Beaufort No.	Name	Wind speed (mean)		State of the sea-surface	Significant wave height, $H_{s,3}$ (m)
		knots	m s^{-1}		
0	Calm	<1	0.0-0.2	Sea like a mirror	0
1	Light air	1-3	0.3-1.5	Ripples with appearance of scales; no foam crests	0.1-0.2
2	Light breeze	4-6	1.6-3.3	Small wavelets; crests have glossy appearance but do not break	0.3-0.5
3	Good breeze	7-10	3.4-5.6	Large wavelets; crests begin to break; scattered white horses	0.6-1.0
4	Moderate breeze	11-16	5.5-7.9	Small waves, becoming longer, fairly frequent white horses	1.5
5	Fresh breeze	17-21	8.0-10.7	Moderate waves taking longer form, many white horses and chance of some spray	2.0
6	Strong breeze	22-27	10.8-13.8	Large waves forming, white foam crests extensive everywhere and spray probable	3.5
7	Near gale	28-33	13.9-17.1	Sea heaps up and white foam from breaking waves begins to be blown in streaks; spindrift begins to be seen	5.0
8	Gale	34-40	17.2-20.7	Moderately high waves of greater length; edges of crests break into spindrift; foam is blown in well-marked streaks	7.5
9	Strong gale	41-47	20.8-24.4	High waves; dense streaks of foam; sea begins to roll; spray may affect visibility	9.5
10	Storm	48-55	24.5-28.4	Very high waves with overhanging crests; sea-surface takes on white appearance as foam in great patches is blown in very dense streaks; rolling of sea is heavy and visibility reduced	12.0
11	Violent storm	56-64	28.5-32.7	Exceptionally high waves; sea covered with long white patches of foam; small and medium sized ships might be lost to view behind waves for long times; visibility further reduced	15.0
12	Hurricane	>64	>32.7	Air filled with foam and spray; sea completely white with driving spray; visibility greatly reduced	>15

WIND WAVE GENERATION



BEAUFORT FORCE 1
WIND SPEED: 1-3 KNOTS

SEA: WAVE HEIGHT .1M (.25FT), RIPPLES WITH THE APPEARANCE OF SCALES, BUT WITHOUT FOAM CRESTS

WIND WAVE GENERATION



BEAUFORT FORCE 4
WIND SPEED: 11-16 KNOTS

SEA: WAVE HEIGHT 1-1.5M (3.5-5FT), SMALL WAVES BECOMING LONGER, FAIRLY FREQUENT WHITE HORSES

WIND WAVE GENERATION



BEAUFORT FORCE 7
WIND SPEED: 28-33 KNOTS

SEA: WAVE HEIGHT 4-5.5M (13.5-19 FT), SEA HEAPS UP, WHITE FOAM FROM BREAKING WAVES BEGINS TO BE BLOWN IN STREAKS ALONG THE WIND DIRECTION

WIND WAVE GENERATION



BEAUFORT FORCE 10
WIND SPEED: 48-55 KNOTS

SEA: WAVE HEIGHT 9-12.5M (29-41FT), VERY HIGH WAVES WITH LONG OVERHANGING CRESTS, THE RESULTING FOAM, IN GREAT PATCHES, IS BLOWN IN DENSE WHITE STREAKS ALONG WIND DIRECTION. ON THE WHOLE, SEA SURFACE TAKES A WHITE APPEARANCE, TUMBLING OF THE SEA IS HEAVY AND SHOCK-LIKE, VISIBILITY AFFECTED.

WIND WAVE GENERATION



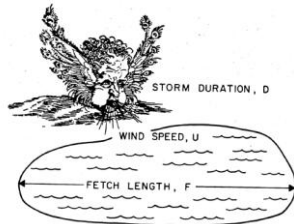
BEAUFORT FORCE 12
WIND SPEED: 64 KNOTS

SEA: SEA COMPLETELY WHITE WITH DRIVING SPRAY, VISIBILITY VERY SERIOUSLY AFFECTED. THE AIR IS FILLED WITH FOAM AND SPRAY

WIND WAVE GENERATION

- Waves are generated by the tangential (shear) and the normal (pressure) stresses exerted by the wind over the sea surface
- Wind energy → wave energy
- Predicting waves by using wind data=wave hindcasting
After world war II

WIND WAVE GENERATION



1. Rate of energy transfer by wind to waves is controlled by wind velocity, u
2. The length of sea area over which the wind blows to create waves, fetch, F
3. Wind duration, t

WIND WAVE GENERATION

Wind Speed - Input for Wave Calculations

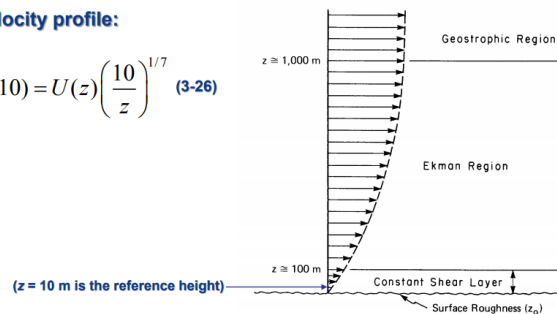
Correction factors for wind measurements:

- gage elevation
- duration
- temperature
- gage location

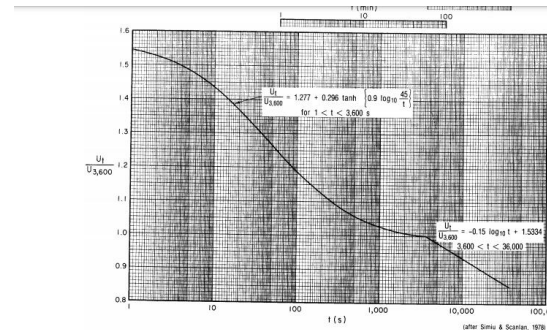
Wind Speed Profile in the Lower Atmosphere

Velocity profile:

$$U(10) = U(z) \left(\frac{10}{z} \right)^{1/7} \quad (3-26)$$



Wind Measurement Duration

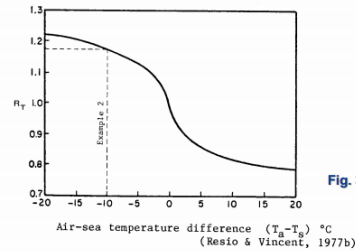


$$U = R_D U(10)$$

Air-Sea Temperature Difference

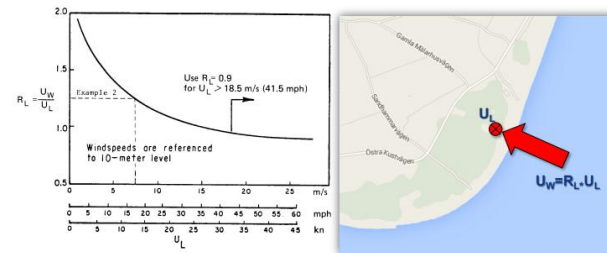
$$U = R_T U(10) \quad (3-27)$$

If temperature measurements do not exist,
 $R_T = 1.1$ should be assumed.



Location of Wind Gage

Ratio between wind speed over water and
 wind speed over land.



WIND WAVE GENERATION

1-Wind speed

- varies with distance above the water
- Standard height = 10m, U_{10}
- Correction: $U_{10}/U_z = (10/z)^{1/7}$, R_z
- Correction due to wind speed over land, R_L
- Correction due to measurement duration R_D
- Correction for air-sea temp difference, R_T

$$U = U_{meas} R_z R_D R_T R_L$$

WIND WAVE GENERATION

1-Wind speed

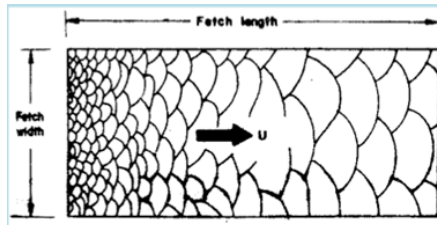
- Wind stress factor U_A (adjusted wind speed)

$$u_A = 0.71 u^{1.23}$$

wave hindcasting

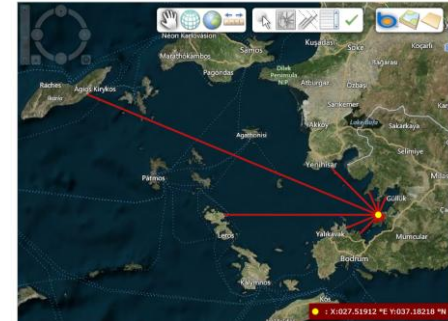
2-Fetch

The sea area over the wind blows to create waves is called the fetch.



wave hindcasting

2-Fetch Calculation?

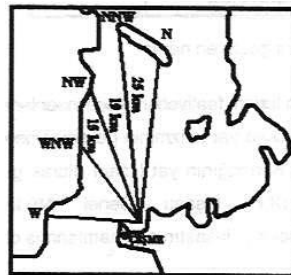


wave hindcasting

2-Fetch Calculation?

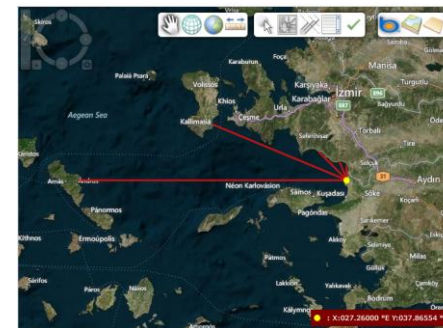
$$F = \frac{\sum F_i \cos^2 \alpha_i}{\sum \cos \alpha_i}$$

± 45 degree
 $\Delta\alpha = 7.5$ degree



wave hindcasting

2-Fetch Calculation?



N 10697.3
NNE 1183.69
NE 261.42
ENE 167.81
E 140.75
ESE 140.34
SE 166.15
SSE 249.38
S 369.1
SSW 611.61
SW 916.55
WSW 1271.03
W 237216.76
WNW 163477.34
NW 35108.11
NNW 18509.8

Saygıyla anıyoruz...



wave hindcasting

3-duration (hrs)

Duration of the wind blowing (storm duration, t) is also an important factor in controlling the growth of waves

For the same U and same F

H and T increase with t

wave hindcasting

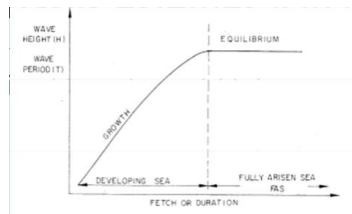
For a given u ,

F, t increase $\rightarrow H, T$ increase

But there is a limit!

Energy rate from wind to wave =

rate of eng dissipation due to wave breaking and turbulence



wave hindcasting

This equilibrium state is called as

Fully developed (arisen) sea FAS: F and t is enough long for u to generate wave with full capacity

Developing sea: F and/or, T not enough long, wave is small

If F is short, fetch limited

If t is short, duration limited

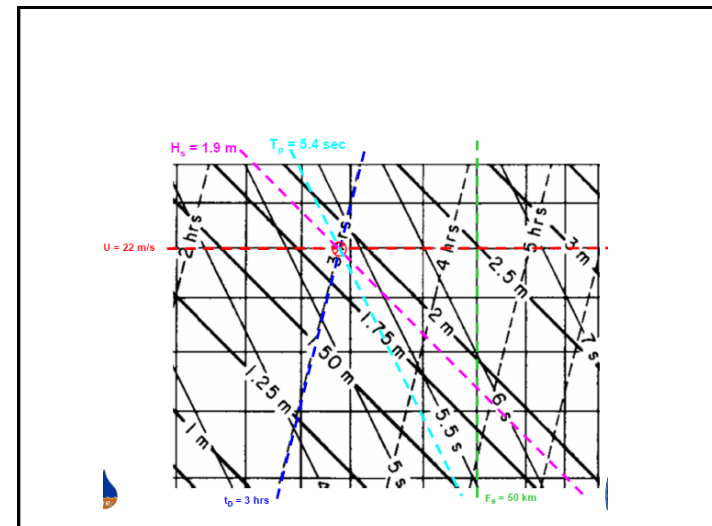
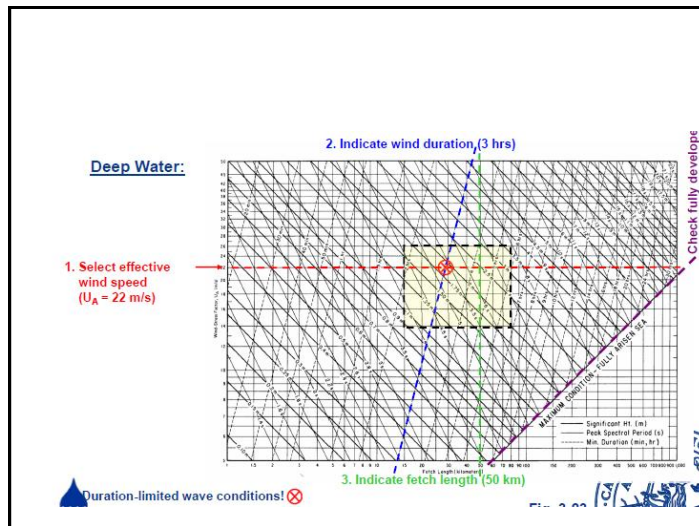
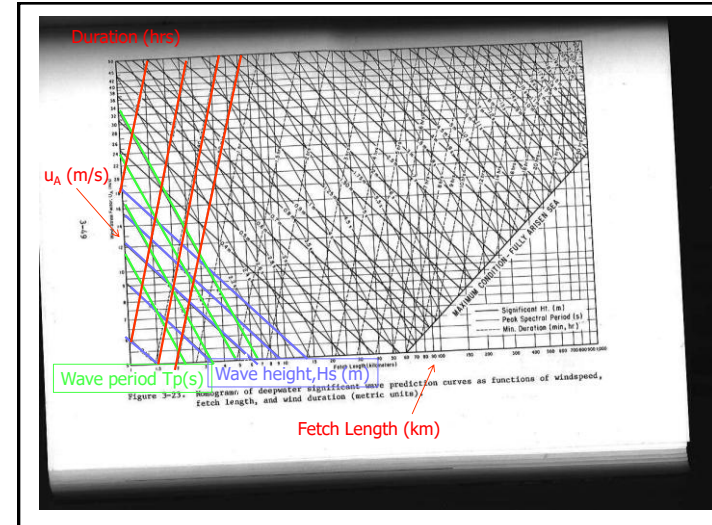
wave hindcasting

Manual models: SMB and JONSWAP

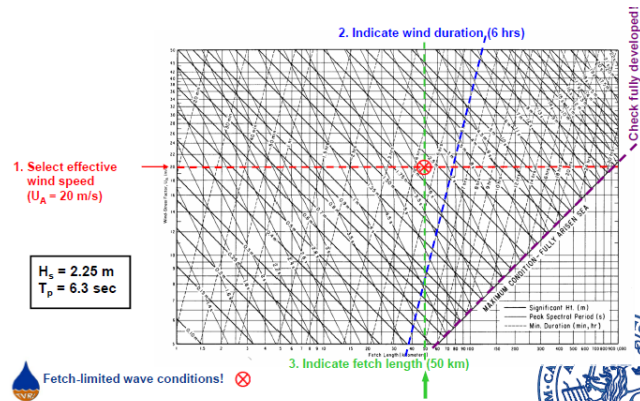
Nomograms

Decide Fully Developed sea or developing sea

If developing, decide fetch or duration limited



Example 1: $U_A = 20$ m/s, $t_D = 6$ hrs, $F_e = 50$ km. $H_s = ?$, $T_p = ?$



wave hindcasting JONSWAP model calculations

Dimensionless parameters:

Fully developed sea:

$$F^* = gF/u_A^2$$

$$H_{m0}^* = 0.243$$

$$H_{m0}^* = gH_{m0}/u_A^2$$

$$T_p^* = gT_p/u_A$$

$$T_p^* = 8.13$$

$$t^* = gt/u_A$$

wave hindcasting

JONSWAP model calculations

Effective fetch (the fetch needed to produce the same wave height if the duration had been infinite)

$$F_{eff}^* = (t^*/68.8)^{3/2}$$

$$F^* < F_{eff}^*$$

Fetch limited

$$F^* > F_{eff}^*$$

Duration limited

$$H_{m0}^* = 0.0016(F^*)^{1/2}$$

Same but put F_{eff}^* instead of F^*

$$T_p^* = 0.286(F^*)^{1/3}$$

Wave hindcasting

Numerical methods

- Concept of energy conservation
- Wind—spectrum— H_{m0} , T_p
- 3G- WAM, SWAN (nonlinear interactions are added)
- More detailed input
- Water depth, wind speed and direction for whole basin as mesh

The sources of wind wave data

Wind measurements (hourly averaged wind speed and direction, daily maximum) by Turkish State Meteorological Office

But stations mostly are on land!!!

The sources of wind data

- numerical model estimates
- commonly used because of their wide temporal and spatial coverage as well as lower cost.

The sources of wind data

Several reanalyses data for climate studies

- ECMWF (European Centre for Medium range Weather Forecast) (Era-Interim)
- NCEP (CFSR)
- Japanese Meteorological Agency (JRA-55)
- NASA (MERRA)

All the reanalysis cover the period from 1979 to the present, except Japanese. In JRA-55 the data starts from 1958.

WAVE DATA

- Wave prediction from wind

Large coverage

Cheap

Wind data ?

Poor result

WAVE DATA OBSERVATION AND MEASUREMENT

- Visual observation
 - ship (DW)
 - coastal stations (in Turkey, DMI, 3/day)
- easy+cheap+long duration and coverage
- Poor accuracy
- Limited information
- Biased data

WAVE DATA OBSERVATION AND MEASUREMENT

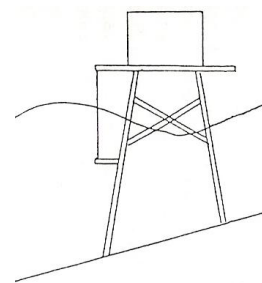
- Instrumental measurement
 - across or on the sea surface (step, resistance gauge, wave buoy)
 - below the surface (pressure)
 - above the surface (laser, radar, remote sensing)
- reliable+accurate+detail information
- Limited duration and coverage
- Expensive

WAVE MEASUREMENT

SOME METHODS FOR WAVE MEASUREMENT

- Instruments measuring waves across or on the sea surface (step, resistance gauge, wave buoy)
- below the surface (pressure, ultrasonic, ADCP)
- above the surface (laser, radar, satellites)

FIXED INSTRUMENT MEASURING WAVE AT THE SEA SURFACE

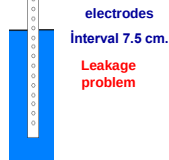


- Generally known as 'WAVE STAFF'
- Mounted on the available structure
- Or build a structure when it is economic
- The output:
 - Can be recorded on site
 - Sent to shore along a cable etc.
- Need to be mounted well away from any sizeable structural members due to reflection

FIXED INSTRUMENT MEASURING WAVE AT THE SEA SURFACE

* Generally known as 'WAVE STAFF'

Step-resistance gauge



Resistance wire staffs



Two Resistance wires

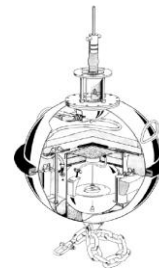
- insulating films form,
- Seaweed,
- refuse material,
- On the surface of wire
- Better for the model experiments but must be cleaned before use



The capacitance between the wire and the sea is measured.

- conducting films form,
- May absorb water

SENSORS IN BUOYS



•Water is too deep or waves too far offshore.

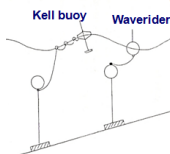
- No need for a cable to shore (wireless antenna and telemetry Hardware)
- solar power can be employed
- Easy deployment
- expensive, easy to handle
- Accurate for small waves

A buoy floating on the sea surface moves up and down with waves.

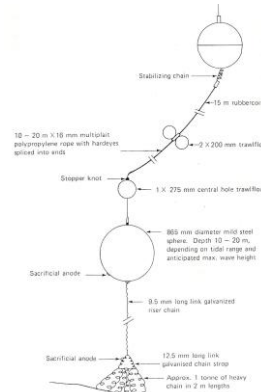
Vertical acceleration is measured.

Integrated twice to give the vertical displacement.

SENSORS IN BUOYS



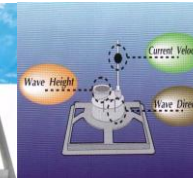
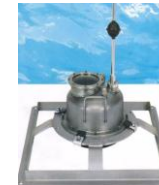
- Must be moored securely,
- Allow them to rise and fall with the highest waves
- Sufficient flexibility
- Position is not fixed in horizontal direction
- Damage from larger waves, vessel traffic or from vandalism
- Errors in accelometer due to pitch and roll



SUB-SURFACE SENSORS

MOUNTED ON OR NEAR THE SEA BED

PRESSURE TYPE SENSORS



•Need a basement(concrete)

•Sea bed condition

Limited high-frequency response

WAVE PRESSURE

$$P = \underbrace{\gamma a \frac{\cosh(k(z+d))}{\cosh(kd)}}_{\text{Dynamic pressure due to water particle motion}} \underbrace{\cos(kx - \sigma t) - \gamma z}_{\text{Hydrostatic}}$$

WAVE PRESSURE

$$P = \gamma a \frac{\cosh(k(z+d))}{\cosh(kd)} \cos(kx - \sigma t) - \gamma z$$

$$K_p = \frac{\cosh(k(z+d))}{\cosh(kd)} \quad \text{Pressure response factor}$$

$$\eta = a \cos(kx - \sigma t)$$

$$P = \gamma K_p \eta - \gamma z$$

WAVE PRESSURE

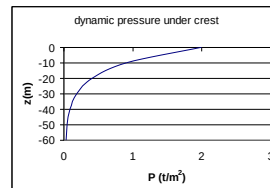
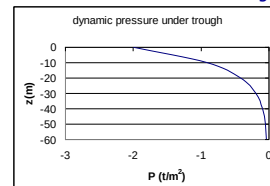
Dyn wave pressure is
Max under wave crest
Min under wave trough

At $z=0$, $K_p=1$, $P=\gamma\eta$

At $z=-d$ $P = \gamma\eta \frac{1}{\cosh(kd)} + \gamma d$

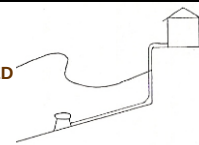
DW ?
 $P=\gamma d$

SW ?
 $P=\gamma(\eta+d)$



SUB-SURFACE SENSORS

MOUNTED ON OR NEAR THE SEA BED



Ultrasonic wave gauge

ADCP (Acoustic Doppler Current Profilers)

Wave and current together

Wave height and direction

Vessel mounted downward-looking is also possible

Horizontal current meter to measure the wave orbital velocities.

ABOVE SURFACE SENSORS

REMOTE SENSING

HF (HIGH FREQUENCY) RADARS

Shore based remote sensing system

EM waves sent to ocean are backscattered on surface waves

Possible to monitor surface currents, waves and winds and direction

Capable of producing map of ocean surface up to 60 200km from shore

Unaffected by weather, clouds, or changing ocean conditions

Not surface elevation but spectrum

ABOVE SURFACE SENSORS

REMOTE SENSING

Satellite Data

There are several satellite instruments relevant to the wind and wave studies.

In particular, Scatterometer, Synthetic Aperture Radar (SAR) and radar altimeter (RA) are of special importance.

Satellites are usually designed to serve for a few years (around 3-7 years) although they last twice as long on average.

ABOVE SURFACE SENSORS

REMOTE SENSING

Satellite Data

Satellite	Launch	End of Life	Altitude	Repeat Cycle
SEASAT	1978	1978	800 km	17 days
Geosat	1985	1990	800 km	17 days
ERS-1	1991	1996	785 km	35 days*
TOPEX/Poseidon	1992	2006	1336 km	10 days
ERS-2	1995	2011 [#]	785 km	35 days*
GFO	1998	2008	800 km	17 days
Jason-1	2001	2013	1336 km	10 days*
ENVISAT	2002	2012	800 km	35 days*
Jason-2	2008		1336 km	10 days
CryoSat-2	2009		720 km	369 days
HY-2	2011		963 km	14 days
SARAL/AltiKa	2013		800 km	35 days

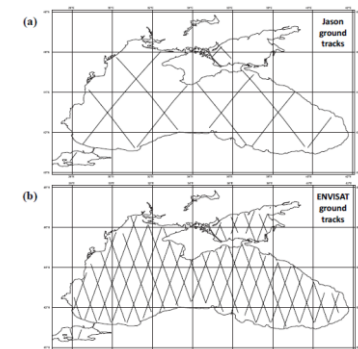
ABOVE SURFACE SENSORS

REMOTE SENSING

Radar Altimeter

The radar altimeter (RA) is a nadir looking active microwave instrument.

This instrument emits pulses and measures the characteristics of the returned signal from the ground. The travel time, the power and the shape of the return pulse are used to estimate the mean sea level, the surface wind speed (u_{10}) and the significant wave height (H_s).



ABOVE SURFACE SENSORS

REMOTE SENSING

Space borne Synthetic Aperture Radar (SAR)

An active microwave instrument producing high-resolution imagery of the Earth's surface

Possible to monitor surface currents, waves, swells and direction, bathymetry changes, air/sea temp changes

Spatial coverage varies 100x100km to 500x500 km

Unaffected by clouds, dusty conditions and solar illumination

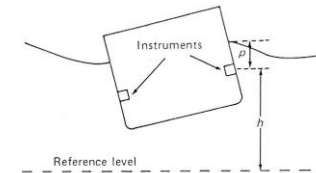
Important for freak waves

SHIPBORNE SYSTEMS

RECORDING WAVES ON THE DEEP SEA DURING A STORM.

MOUNTED IN WEATHER SHIPS AND LIGHT VESSELS.

Shipborne systems



The principle of the Shipborne Wave Recorder. The sum of the pressure p and the twice-integrated vertical acceleration h gives the estimate of sea-surface elevation.

Two accelerometers sensor boxes are mounted symmetrically on each side of the ship and pressure sensor connected.

The accelerometer outputs are integrated twice and added to the pressure signal, giving in each case the surface elevation relative to a fixed horizontal plane.

Average of those are giving in concept the surface elevation at the centre of the ship.

Conclusion on wave measurement

- Each of the instrumentation/application has its own advantages and disadvantages
- Try to get detailed knowledge
- Choose according to needs, importance, cost, duration of the project

Wave measurement in Turkey

- NATO TU Waves Project by METU, Medcoast (94-96, Hopa:94-99, not continuous)
- Wave rider buoy



Wave measurement in Turkey

- Ministry of Transport, DLH Gen. Dir. Research Dept
- Pressure type and ultrasonic type
- Filyos (1994-1996)
- Çandarlı (2001)
- Ambarlı (2002)
- Karaburun (2003)

Wave measurement in Turkey

STATE METEOROLOGICAL ORGANISATION (2015-)

- Silivri
- Karaburun
- Çanakkale
- Antalya
- Mersin (silifke)